Ethical Co-Development of AI Applications with Indigenous Communities

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1 Abstract/Course Description

This course explores how researchers and practitioners can engage ethically with Indigenous communities when developing AI- and data-intensive applications. Some key issues such as fair engagement, legal constraints, reciprocity, and informed consent are discussed based on the examples draw from the instructors' experience. The course also examines good practices in terms of co-designing and co-development processes, data governance and sovereignty issues and systems, decolonial software licensing, and processes of technology transfer and appropriation. In its practical part, the course critically discusses examples and cases gathered from the audience to explore the diversity of issues and solutions when working with Indigenous communities.

2 Benefits

The course attendees are expected to gain a broad but introductory understanding of key issues and ethical practices that they should be aware and consider when engaging with Indigenous communities. This knowledge should help them to avoid common pitfalls when considering and planning engagements with such communities and, at the same time, support critical evaluations of ongoing engagement processes and of previous research. The course also covers design and evaluation methods which have been used in Indigenous contexts and discusses how HCI practitioners can co-design technology with Indigenous communities using culturally-appropriate methods and decolonial perspectives.

3 Intended Audience(s)

This course is intended to a broad audience of researchers and practitioners of HCI, development, design, and social impact projects, who are considering or engaged in application design and development with Indigenous communities. The course also looks into the similarities and differences of working with vulnerable communities, so it may also be of interest for attendees involved in other types of community-related application development. The course is taught from the perspective of HCI and AI practitioners who have engaged with Indigenous communities and, in the process, learned the hard way how to conduct such collaborations ethically, respectfully, and effectively.

4 Prerequisites

The attendees of the course are expected to have a general knowledge of HCI, design, and development practices, including traditional design and evaluation methods. Some basic knowledge of AI applications is also helpful to more easily understand the data-related issues which are common in cross-cultural and community-focused engagements.

5 Content

1. **Introduction and Motivation** (10 min)
2. **Indigenous Communities, Languages, and Cultures** (15 min)
3. **Partnering with Indigenous Communities** (20 min)
   - 3.1 Principles and guidelines
   - 3.2 Colonial and decolonial perspectives
   - 3.3 Legal perspectives and constraints
   - 3.4 Relational accountability
   - 3.5 Case: Partnering with FOIRN in Alto Rio Negro, Brazil
4. **Data Collection** (20 min)
   - 4.1 Informed consent
   - 4.2 Data control and sovereignty
   - 4.3 Intellectual property
   - 4.4 Case: the Māori license *Kaitiakitanga*, New Zealand
5. **Discussion** (10 min)
   BREAK
6. **Summary of Previous Module** (5min)
7. **Co-Designing Applications** (15 min)
   - 7.1 Cultural considerations
   - 7.2 Design methods and processes
   - 7.3 Being vs. Doing the Indigenous way
   - 7.4 Case: Designing with high school students in the Tenondé-Porã community, Brazil
8. **Co-Developing Applications** (15 min)
   - 8.1 Problems with traditional development models
   - 8.2 Usage-focused development models
   - 8.3 Support, maintenance, and continuity

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8.4 Technology transfer and appropriation
8.5 Case: deploying a writing assistant to Nheengatu speakers, Brazil
9. **Practical Work: Designing an Engagement** (30 min)
9.1 Gathering cases from audience
9.2 Group work: identifying issues
9.3 Group work: making recomendations
9.4 Presentations by each group with comments
9.5 General discussion
10. **Final Discussion** (10 min)

6 Practical Work

This course includes a practical activity where engagement cases brought by the attendees are discussed and analyzed by the instructors and the audience, as a way to make the presented concepts and ideas more concrete. The instructors will collect from the attendees cases of engagement with Indigenous communities (and provide them, if needed), and, in groups, the attendees will discuss and suggest suitable engagement approaches, proposals, and methodology. The groups will then present to all the attendees and the instructors will coordinate a discussion about the proposed ideas.

7 Instructors Background

Claudio Pinhanez is a scientist, innovator, and professor. He is currently a Principal Scientist in the laboratory of IBM Research in Brazil where he leads research on artificial intelligence, human-machine interaction, and natural language processing. He is also the Deputy Director of the Center for Artificial Intelligence of the University of São Paulo, where he is also a Visiting Professor at the Institute of Advanced Studies. Claudio got his Ph.D. from the MIT Media Laboratory in 1999, joined IBM Research the same year, and in 2010 co-founded its Brazil laboratory. Claudio has more than 150 papers published in journals and scientific conferences and more than 30 patents issued in the USA, Europe, and Japan. He is a Senior Member of ACM and an associate editor of 4 scientific journals. Since 2022 he leads a joint project of IBM Research and the University of São Paulo focused on the use of AI technology to document and vitalize Brazilian Indigenous languages[1]. Recent related publications include a paper at IJCAI 2023 on ethical engagement with Indigenous communities [28], a short paper at a CHI 2023 workshop on designing in such contexts [29], a comprehensive description of his group's research with Indigenous communities [27], besides more technical related research [3, 5, 26, 36]. His work with Indigenous communities has also been recently featured in one of the keynote speeches of NAACL 2024[2].

Edem Wornyo, an engineer, inventor, and senior strategy servant leader, has guided and collaborated with diverse teams of software engineers, UX researchers, program managers, product managers, and executives to establish strategic directions for various products. Currently, he directs the non-extractive data artificial intelligence (AI) research program at Google Research in New York. In his research, Wornyo works closely with Indigenous communities worldwide, focusing on ethical engagements, natural language processing, and data sovereignty. With 14 patent applications to his name, Wornyo is also an accomplished inventor. His research has been published in renowned journals. Wornyo holds a doctorate in Electrical and Computer Engineering with a minor in Materials Science and Engineering from Georgia Tech, USA. His PhD research explored the application of neural networks for information storage in smart materials. Additionally, he has earned a certificate from the Harvard Business School in AI Data Analytics.

The instructors recently co-organized, with others, the first ever ACM FAccT CRAFT session on "Responsible Non-extractive AI Research and Collaboration with Indigenous Communities" [3].

8 Examples of Topics Discussed in the Course

Trust building: Communities around the world have developed distrust as a result of centuries of oppression meted out to these marginalized communities. The literature has delved into various reasons why these communities have this distrust towards corporate bodies [4, 11].

In working with one community, some members of the community have expressed their misgivings about collaborating with us. Some of the reasons given were that the community members do not feel comfortable working with corporate bodies because of previous extractive experiences that some community members suffered when corporate bodies came to them asking for collaboration, but later exploiting the communities and leaving these communities worse than they have found them.

The difficulties we faced stemmed from the initial distrust and skepticism voiced by these historically marginalized communities. Their past experiences had understandably fostered a sense of wariness towards outsiders. However, through our consistent actions and demonstrated empathy towards their plight, we were gradually able to break down these barriers.

Benefits: Indigenous communities, due to historical and ongoing systemic injustices, frequently receive inadequate budget allocation and support from their respective nation-states and territories [6, 23, 38]. This chronic underfunding perpetuates a cycle of impoverishment and severely limits their opportunities for advancement in critical areas such as education, healthcare, infrastructure, and economic development [6, 7].

As a result, these communities often lag significantly behind neighboring non-Indigenous communities in terms of socioeconomic indicators, quality of life, and access to essential services. This disparity not only undermines the well-being and potential of Indigenous peoples but also represents a failure of the state to uphold its obligations and responsibilities towards its Indigenous citizens [13, 18, 24, 25]. For centuries, marginalized communities have endured exploitation and systemic oppression. This historical injustice necessitates a deliberate and conscientious effort to engage with these communities, fostering collaboration and empowering them to reclaim their autonomy.

Central to this endeavor is establishing meaningful dialogues, actively listening to community members, and prioritizing their needs

[1] https://c4ai.inova.usp.br/research_2/#ProIndL_B_eng
[2] https://2024.naacl.org/program/keynotes/
[3] https://programs.sigchi.org/facct/2024/program/content/165874

and aspirations. By working hand-in-hand, we can support initiatives which revitalize and strengthen these communities. This may include, for instance, collaborative efforts to preserve and promote Indigenous languages that have been marginalized and endangered by dominant colonial languages. Through such concerted action, we can begin to redress historical wrongs and forge a path towards a more equitable and just future.

Ethical and Legal Issues: Doing research and development with Indigenous peoples is subjected to specific guidelines and legal constraints. Mihesuah [19] provides a comprehensive set of guidelines for research with US American Indigenous communities. Similarly, Straits et al. [33] proposed engaging in research with Indigenous communities based on 11 principles, including native-centrism, co-learning and ownership, continual dialogue, transparency, accountability, integrity, and community relevance.

In fact, there is a lot of distrust from many Indigenous communities towards researchers and academic work, resulting from a history of exploitation, disregard, and knowledge extractivism [32]. To address these issues, Smith [32] proposes that *relational accountability* [39] should guide these engagements since it is inherent to Indigenous ways of doing. The key idea behind relational accountability is that relationships with the communities are important in research and that the parties are responsible for maintaining them.

There are also legal and regulatory procedures that must be followed when working with specific Indigenous communities. For instance, research, in such cases, should include special procedures for informed consent processes and the involvement of community members in defining exposure and risk to the community. Llanes-Ortiz [17] has proposed a research framework based on 7 key approaches for digital initiatives with Indigenous languages, based on: facilitating digital communication; multiplying content online; normalizing the use of Indigenous languages online; educating in and teaching online; reclaiming and revitalizing digitally; imagining and creating new digital media; defending spaces for Indigenous languages and linguistic rights; and protecting heritage and communities.

Data Sovereignty: Particularly relevant to the developing of AI-based tools are issues related to data sovereignty, consent, and intellectual property (IP) rights [9]. When it comes to AI-related research, gathering, controlling, and using linguistic data become essential parts of the research process and, therefore, of ethical concerns. The sovereignty of Indigenous data has attracted considerable discussion in recent years, particularly in the case of population and genetic data [14, 37], but also in digital and AI research [17]. There are also some efforts to create new agreements and practices, such as the *Kaitiakitanga* data license[4] proposed by members of the *Māori* language community (see also the links provided by *TeHiku*[5]). A more specific guide for AI-related work, including tool and technology design methodologies, was proposed by the *The Indigenous Protocol and Artificial Intelligence (A.I.) Working Group* [16].

[4]https://github.com/TeHikuMedia/Kaitiakitanga-License
[5]https://tehiku.nz/te-hiku-tech/te-hiku-dev-korero/25141/data-sovereignty-and-the-kaitiakitanga-license.

Co-Designing and Co-Development: Regarding the design process, as discussed in [29], the last 15 years have produced many scholarly works about engaging with Indigenous communities for the design of digital artifacts [2, 20–22, 30, 31, 34, 35]. Most of these works focus on describing key principles that designers should follow when engaging with Indigenous communities. However, while this literature offer extensive descriptions of **what** should be done in the design process, it often provides very limited on **how** the actual design process should happen. There are a few exceptions were details about how design workshops were conducted were actually provided [21, 34, 35]. However, even in those cases a systematic description of the methods and means was never provided nor good ideas and practices on how to actually conduct the design workshops.

Interestingly, some works discussing community research related to Indigenous education, schools, and connectivity [1, 8, 10, 15] and Indigenous writing [8, 12] are more helpful. In particular, the discussion in [1] about **being** an Indigenous person vs. **doing** the Indigenous way has helped us in some of the work we have done with Indigenous communities [27] and is discussed in the course.

9 Accessibility

No special provisions beyond those provided by the conference.

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